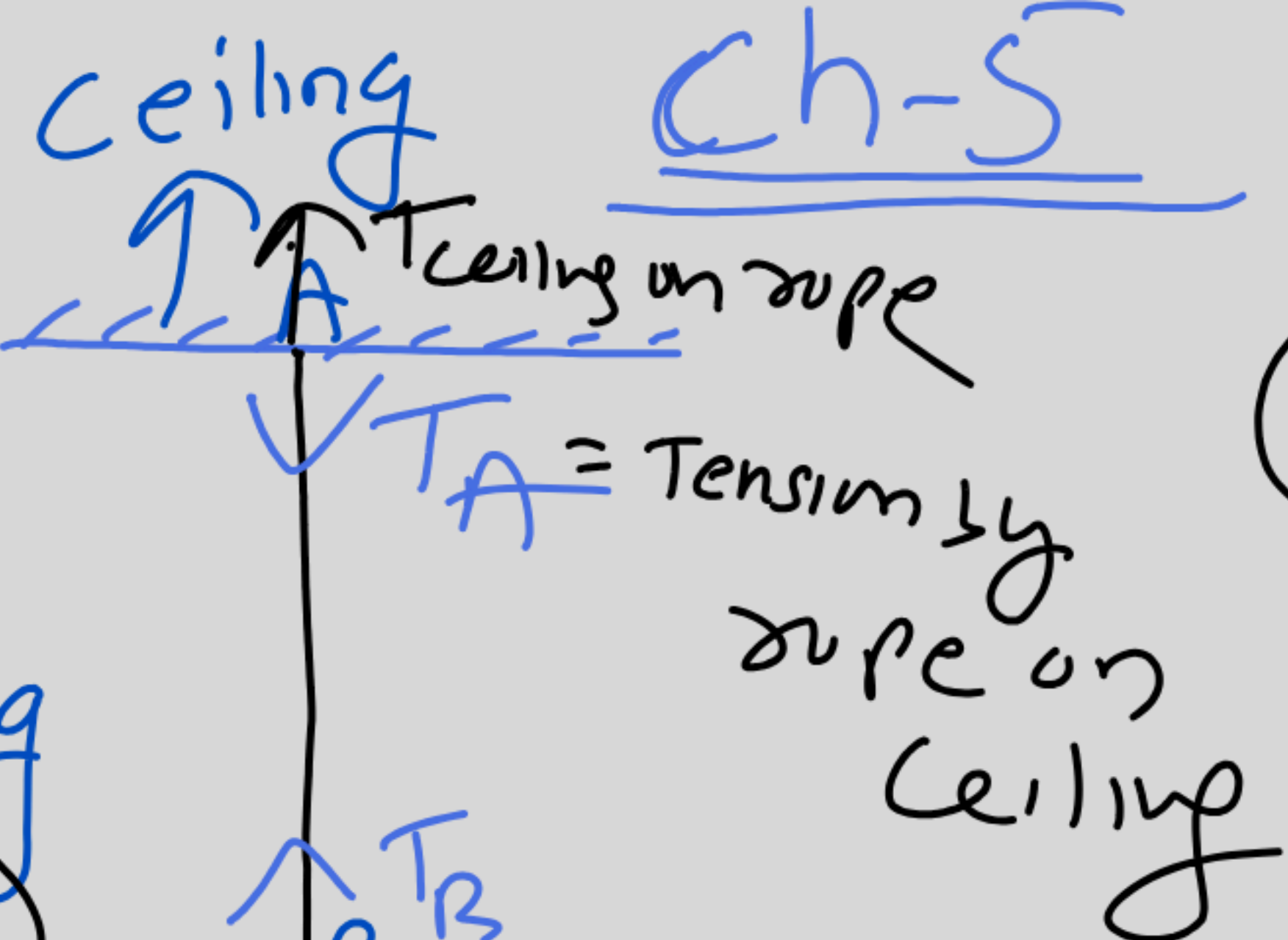


ceiling Ch-5 Application of NLM

Ex 5.1

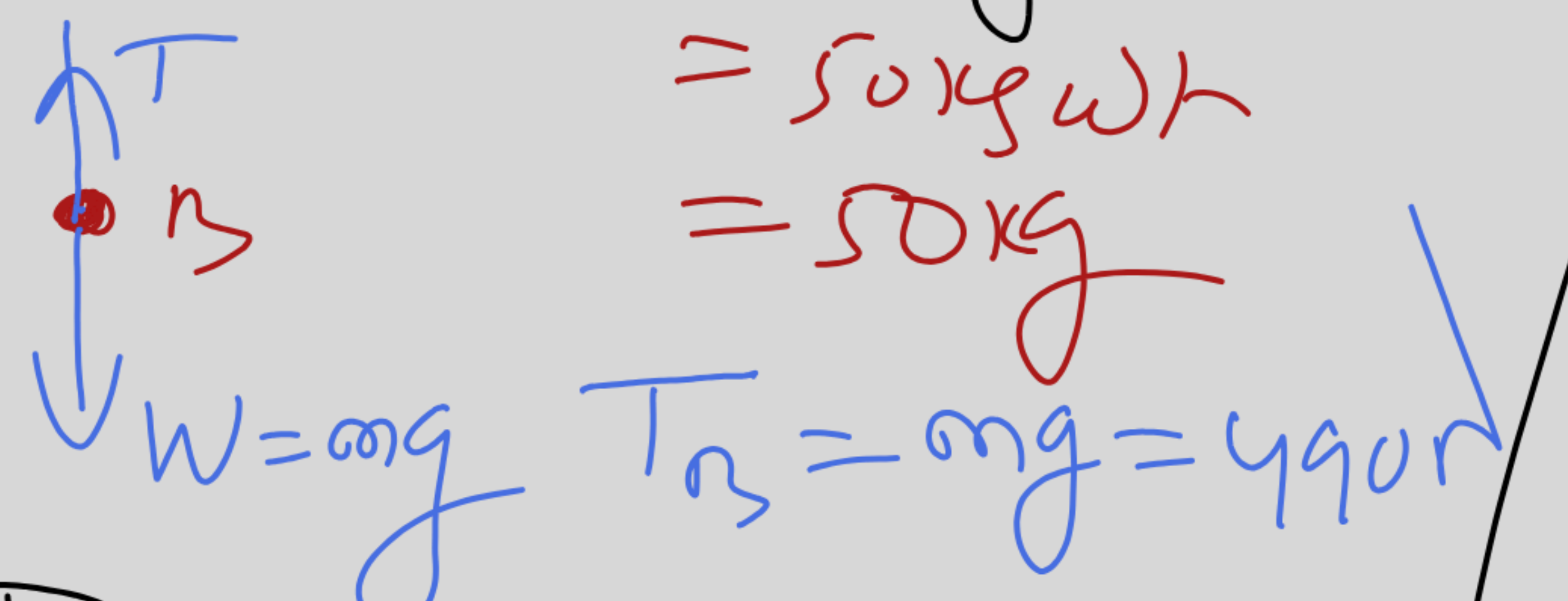
Mass of man = 50 kg
rope is light (massless)

- Find (a) Wt of man
(b) Force by man on rope
(c) Force (Tension) ceiling on rope:



(a) $W = mg = 50 \text{ kg} \times 9.8 \text{ m/s}^2 = 490 \text{ N} = 50 \text{ kgf} = 50 \text{ kg wt} = 50 \text{ kg}$

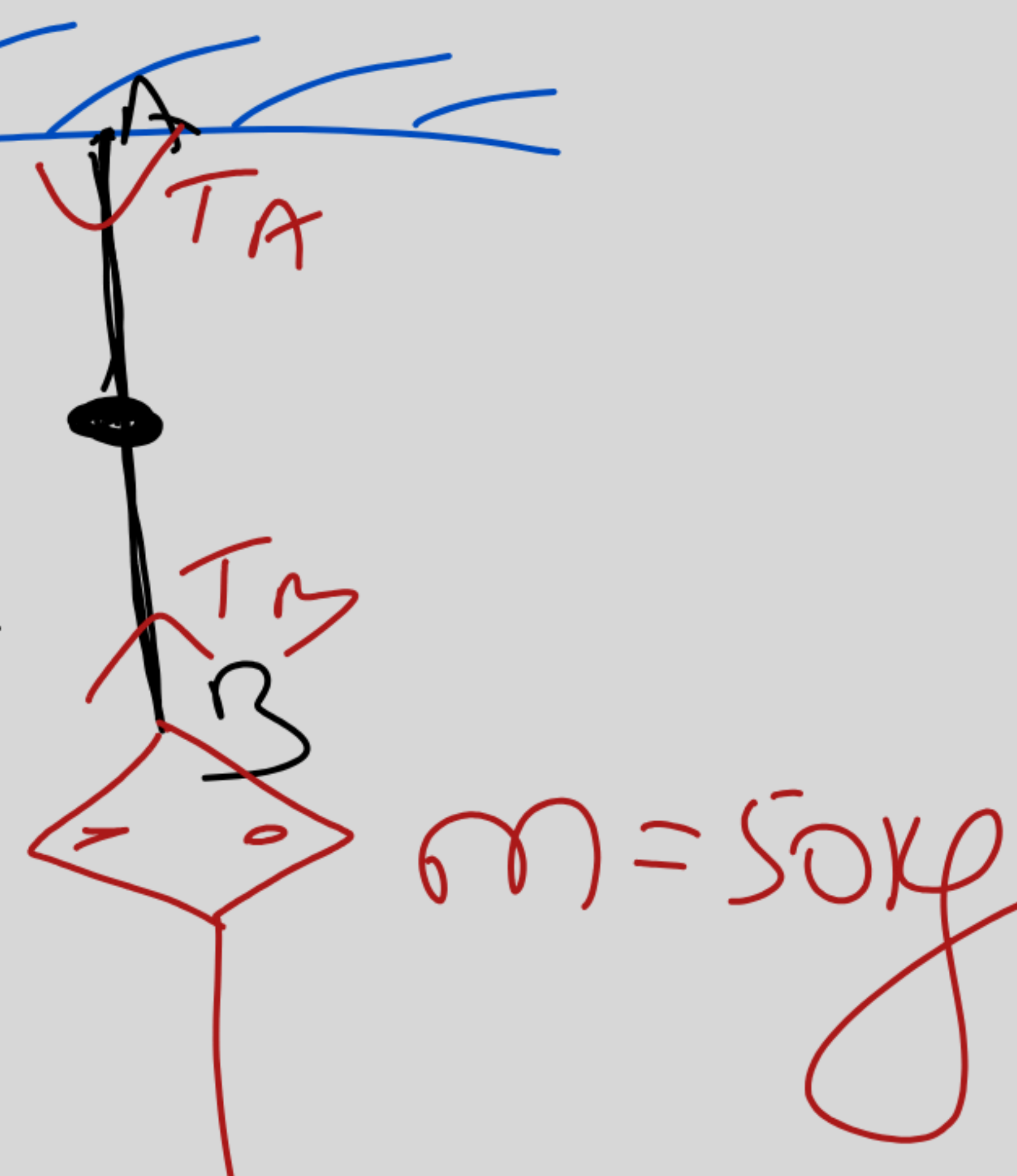
(b)



(c) $T_A = T_B = 490 \text{ N}$

Ex 5.2

$W_{AB} = 1200$
 $m_{AB} = \frac{1200}{9.8} \text{ kg}$



(b)

$T_B = 490 \text{ N}$

$T_{\text{ceiling on system}}$

$T_A = W_{\text{man}} + W_{\text{rope}} = 490 + 1200 = 1690 \text{ N}$

Ex 5.5

Find the tension on the rope.

$T = W_2$ — (1)

Force components on the block on the incline:

- Normal force: $N = W_1 \cos 15^\circ$
- Tension: $T = W_1 \sin 15^\circ$
- Weight: $T = W_2$

Equating the two expressions for tension:

$$W_1 \sin 15^\circ = W_2$$

$$\Rightarrow W_1 \sin(15^\circ) = W_2$$

$$\Rightarrow W_2 = 0.26 W_1$$

Ex 5.8 Q.4 (b) Oct '2006

An elevator & its load = 800 kg coming to stop after covering 25 m with const acc.

Initial velocity $u = 10 \text{ m/s}$

Distance covered $s = 25 \text{ m}$

Final velocity $v = 0$

Acceleration a

Find Tension in cable of elevator when it is being brought to rest

$$v^2 - u^2 = 2as$$

$$\Rightarrow 0^2 - 10^2 = 2a \times 25 \Rightarrow a = \frac{-100}{50} = -2 \text{ m/s}^2$$

or

$$0^2 - (-10)^2 = 2a(-25)$$

$$\Rightarrow -10^2 = -2a \times 25$$

$$a = 2 \text{ m/s}^2$$

$$T - W = ma$$

$$\Rightarrow T = W + ma = mg + ma = m(g + a)$$

$$= 800(9.8 + 2) = 9440 \text{ N}$$

Ty U 5.1

92.67%

Force components:

- Horizontal: $T \cos 45^\circ$
- Vertical: $T \sin 45^\circ$

Equilibrium equations:

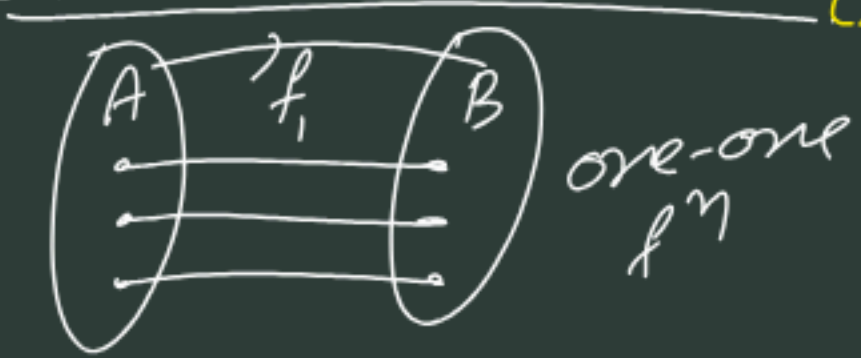
$$2T \cos 45^\circ = W$$

$$2T \times \frac{1}{\sqrt{2}} = W$$

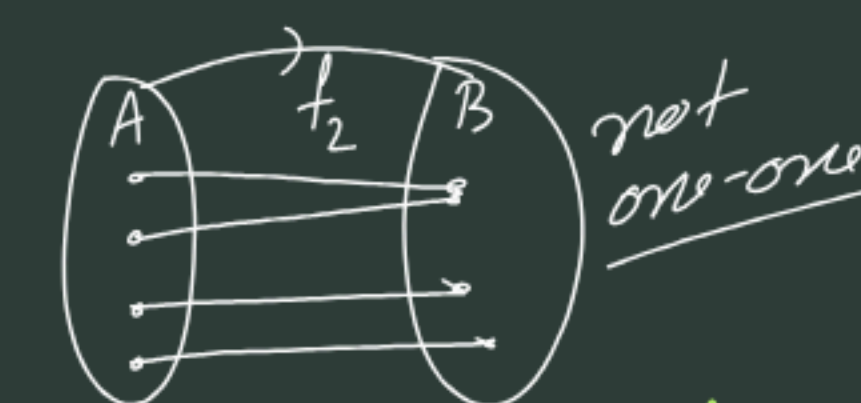
$$\Rightarrow T = \frac{W}{\sqrt{2}}$$

- T in cable
- (a) $\frac{W}{2}$ (b) $\frac{W}{\sqrt{2}}$
- (c) $W\sqrt{2}$ (d) $2W$

one-one function (Injective f.m)



one-one
 f_1



not one-one
 f_2

let f be a function from A to B .

f is one-one function iff

$\forall x_1, x_2 \in A$, if $f(x_1) = f(x_2)$ then $x_1 = x_2$ ✓

ex $f(m, n) = 2m - n$
 $f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$

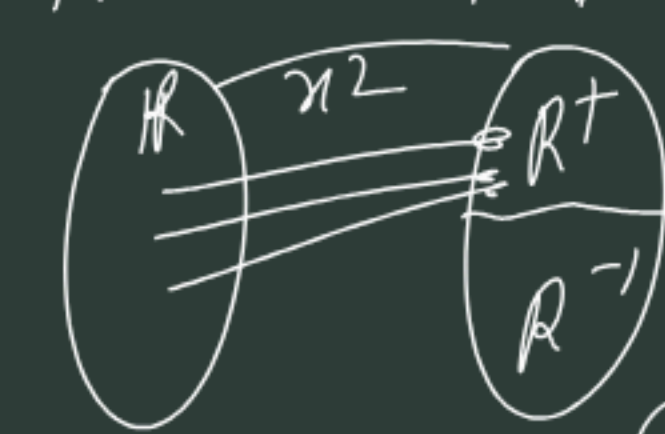
sl $f(m, n) = 2m - n$

$$f(1, 1) = 2 - 1 = 1$$

$$f(3, 5) = 2 \cdot 3 - 5 = 1$$

f is not one-one.

$f(x) = x^2$ $f: \mathbb{R} \rightarrow \mathbb{R}$



co-domain = $\mathbb{R}$
Range = $\mathbb{R}^+$

ex $f(x) = 3x + 7$ $f: \mathbb{R} \rightarrow \mathbb{R}$

sl $\forall x_1, x_2 \in \mathbb{R}$

$$\text{let } f(x_1) = f(x_2)$$

$$3x_1 + 7 = 3x_2 + 7$$

$$x_1 = x_2$$

f is one-one.

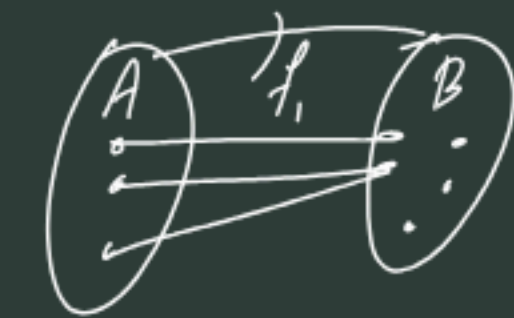
ex $f(x) = 3x^2 + 7$ $f: \mathbb{R} \rightarrow \mathbb{R}$

sl $x_1 = 1, x_2 = -1$

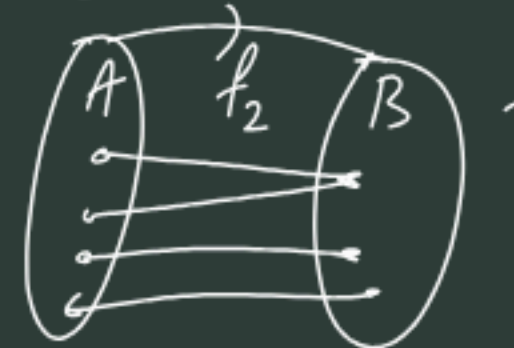
$$f(1) = 10, f(x_2) = 10$$

f is not one-one

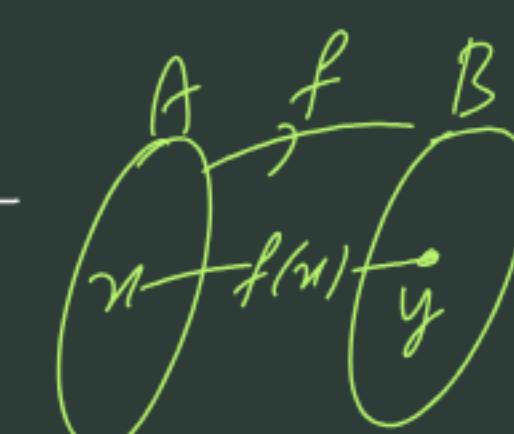
onto functions (Surjective)



f_1 is not onto.



f_2 is onto.



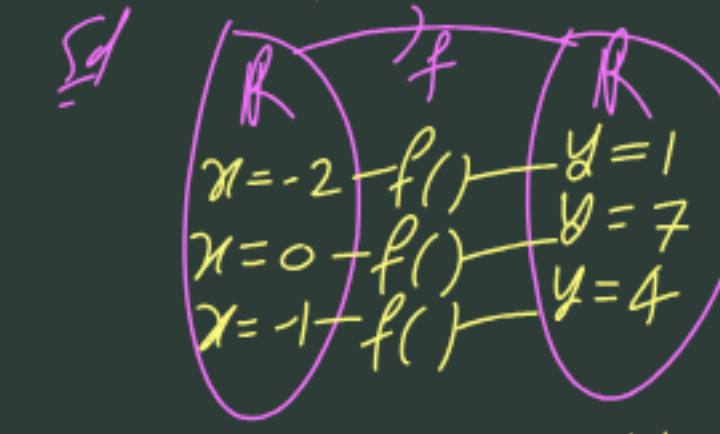
f is a onto function from

A to B iff $\forall y \in B$ (co-domain)

$\exists x \in A$ (domain) such that

$$f(x) = y$$

ex $f(x) = 3x + 7$
 $f: \mathbb{R} \rightarrow \mathbb{R}$



$$x = \frac{y-7}{3}$$

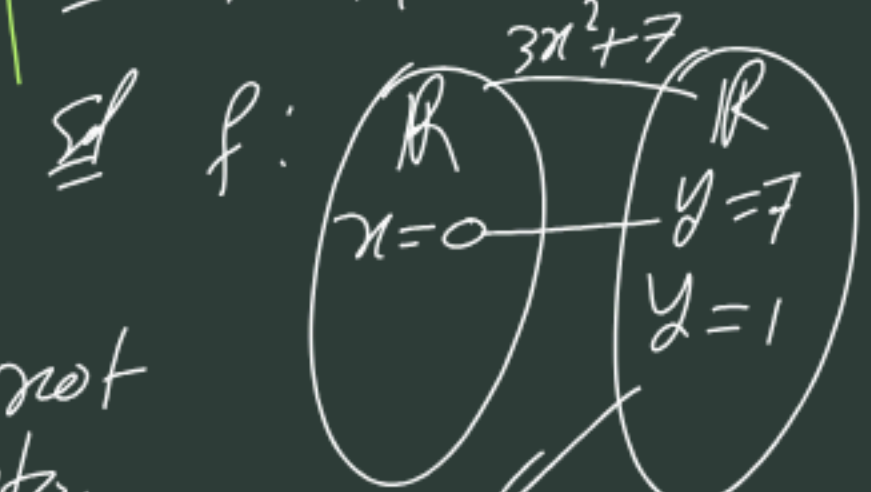
$$\begin{cases} 3x + 7 = y \\ x = \frac{y-7}{3} \end{cases}$$

$$\forall y \in \mathbb{R}, \exists x = \frac{y-7}{3} \in \mathbb{R}$$

$$\text{sl } f(x) = f\left(\frac{y-7}{3}\right) = \left(\frac{y-7}{3}\right)3 + 7 = y$$

f is onto f^{-1}

ex $f: \mathbb{R} \rightarrow \mathbb{R}$ $f(x) = 3x^2 + 7$



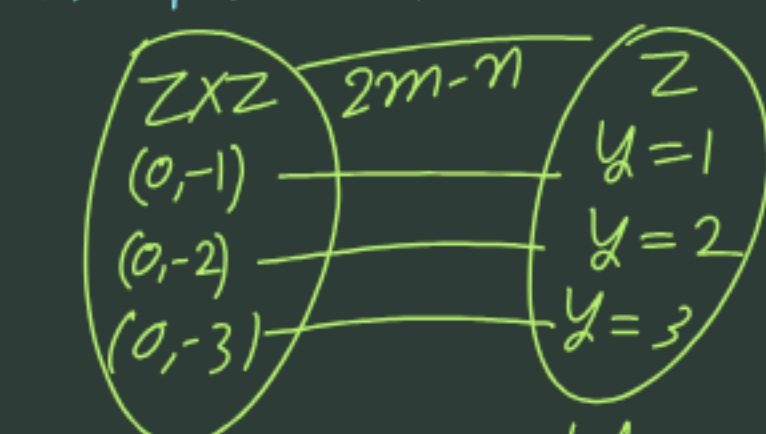
f is not onto.

$$\frac{\sqrt{1-7}}{3} = \frac{\sqrt{-6}}{3} \text{ complex}$$

$$\begin{cases} 3x^2 + 7 = y \\ x = \frac{\sqrt{y-7}}{3} \end{cases}$$

ex $f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$

(a) $f(m, n) = 2m - n$



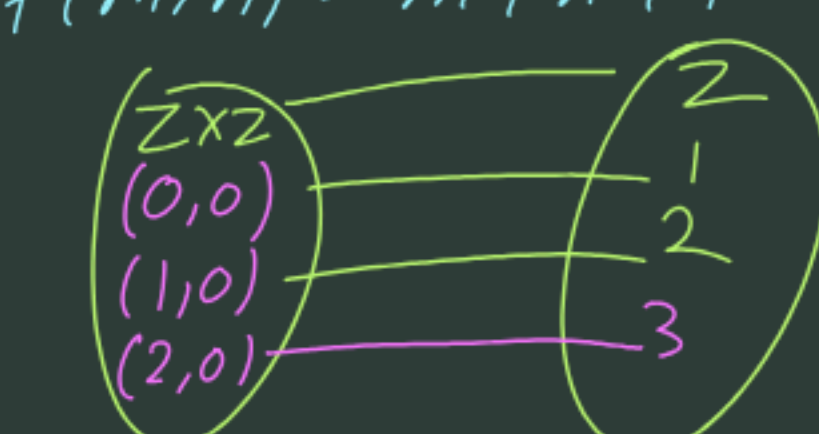
$$\begin{pmatrix} 0, -y \\ y, y \end{pmatrix} \quad y$$

$$\forall y \in \mathbb{Z}, \exists (m, n) = (0, -y) \in \mathbb{Z} \times \mathbb{Z}$$

$$\text{sl } f(m, n) = f(0, -y) = 2 \cdot 0 - (-y) = y$$

f is onto.

(b) $f(m, n) = m + n + 1$



$$\forall y \in \mathbb{Z}$$

$$\exists (m, n) = (y-1, 0)$$

f is onto.

>> Signed Right Shift.

int a = -15;

int b = a >> 2;

so.pln(b);

-15 >> 2

$$\begin{array}{r} 15 \Rightarrow \begin{array}{ccccccc} 0000 & \dots & 1111 \\ 1111 & \dots & 0000 \\ \hline & & +1 \end{array} \\ -15 = \begin{array}{ccccccc} 1111 & \dots & 0001 \end{array} \end{array}$$

$$\Rightarrow \begin{array}{ccccccc} 1111 & \dots & 0001 \end{array}$$

$$\begin{array}{r} \boxed{1}111 \dots 1100 \\ 000 \dots 0011 \\ \hline \boxed{1}000 \dots 0100 \\ \textcircled{-4} \end{array}$$

>>> (Unsigned Right Shift)

int a = -15;

int b = a >>> 2;

s.o.pln(b);

→ In -ve no

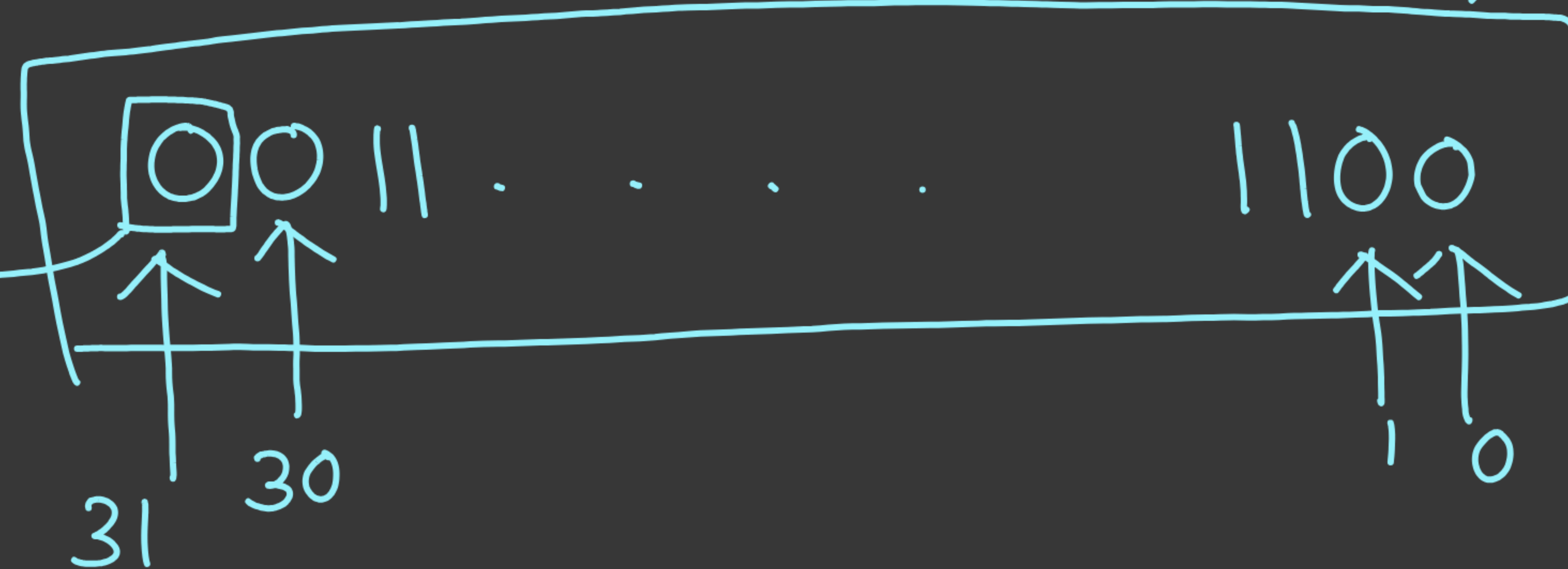
RHS → bits ll be discarded

LHS → 0's ll be padded.

-15 ⇒ 001111...0001

01111111...111
↓
2147483647

Signbit
+ve



$$2147483647 - 2^{30} = 3$$
$$= 1073741820$$

int a = -5;

int b = a >>> 1;

sizeof(b);

2147483645

-5 $\Rightarrow$ 01111 1011



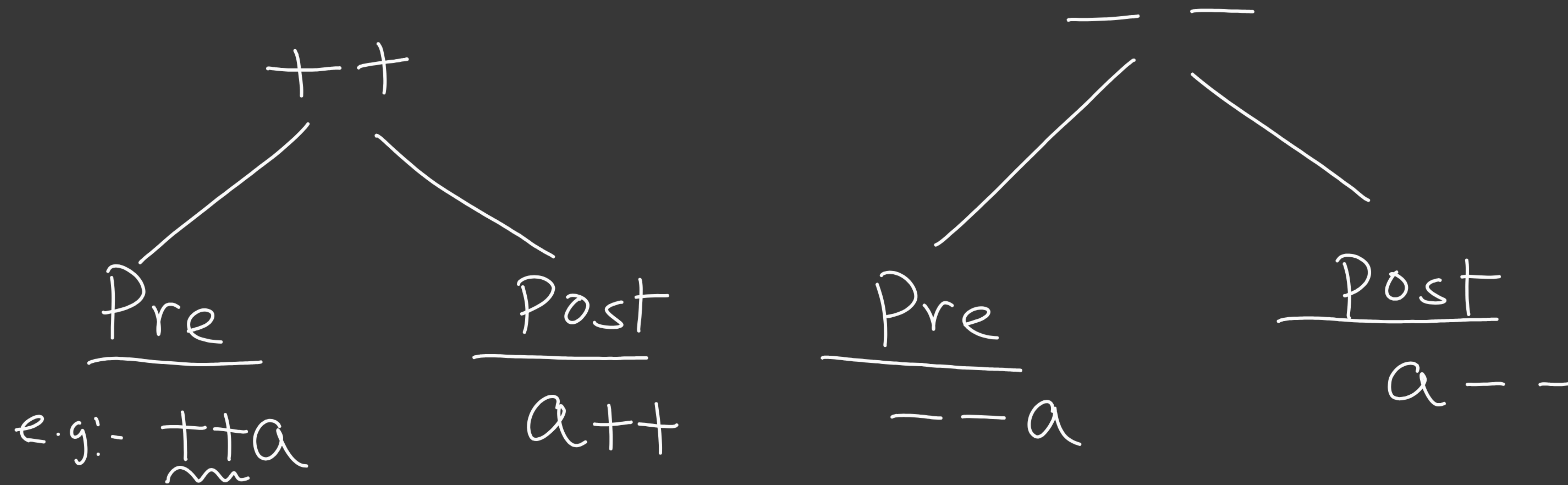
01111

1101

↑
2' = 2

2147483647 - 2

Increment and Decrement Operator.
(++) (--) :- Unary Operator.
↓
one operand.



$++a/a++ \Rightarrow a = a + 1$

$--a/a-- \Rightarrow a = a - 1$

① int a=5;
++a;
s.o.pln(a);

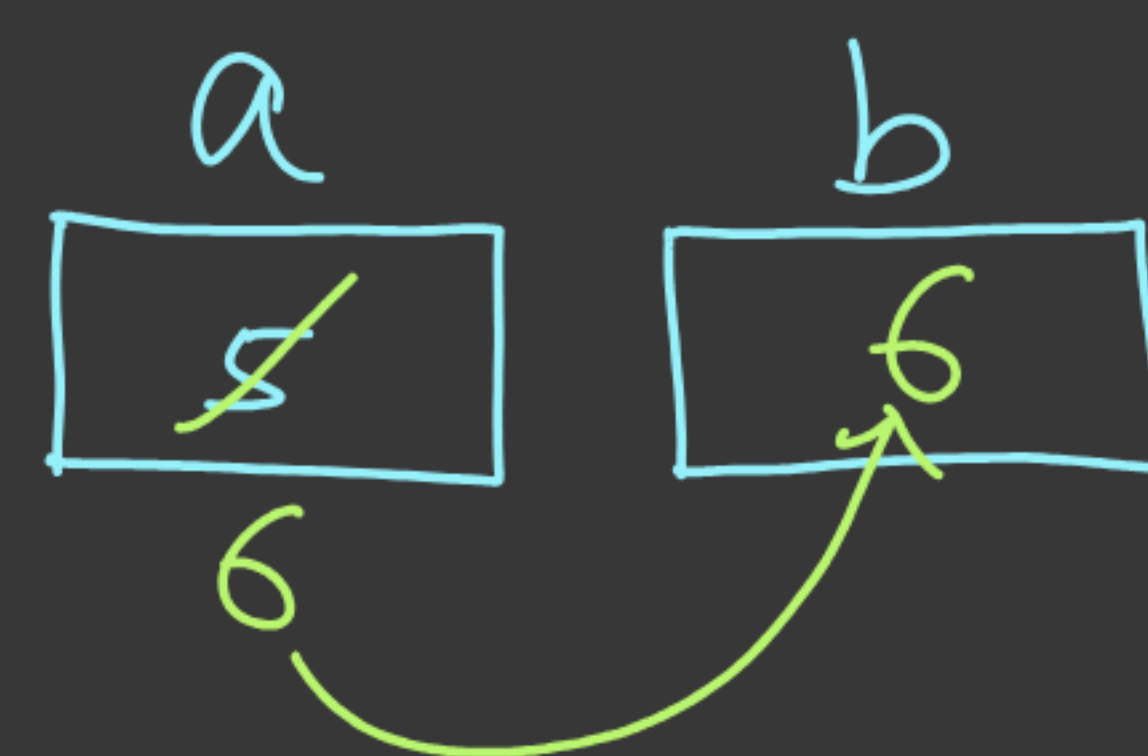
② int a=5;
a++;
s.o.pln(a);

③ int a=5;
--a;
s.o.pln(a);

④ int a=5;
a--;
s.o.pln(a);

int a=5;
int b=++a;
s.o.pln(a+" "+b);

① Assignment (2nd)
② increment (1st) pre



int a=5;
int b=a++;
s.o.pln(a+" "+b);

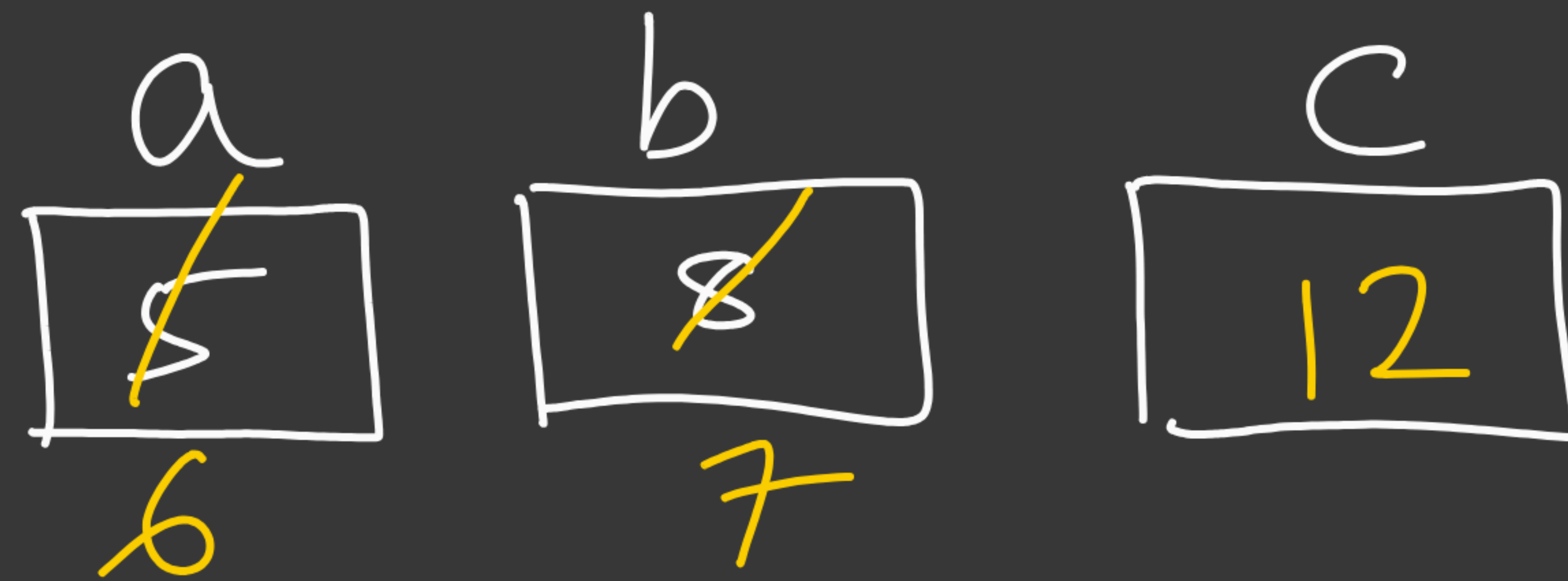
① Assign (1st)
② Increment (2nd) (post)

```

int a=5, b=8;
int c = 5a++ + --b;
s.opln(12(a + "" + b + "" + c),
      6      7      12

```

$a = a + 1$
 $6 = 5 + 1$
 $b = b - 1$
 $7 = 8 - 1$



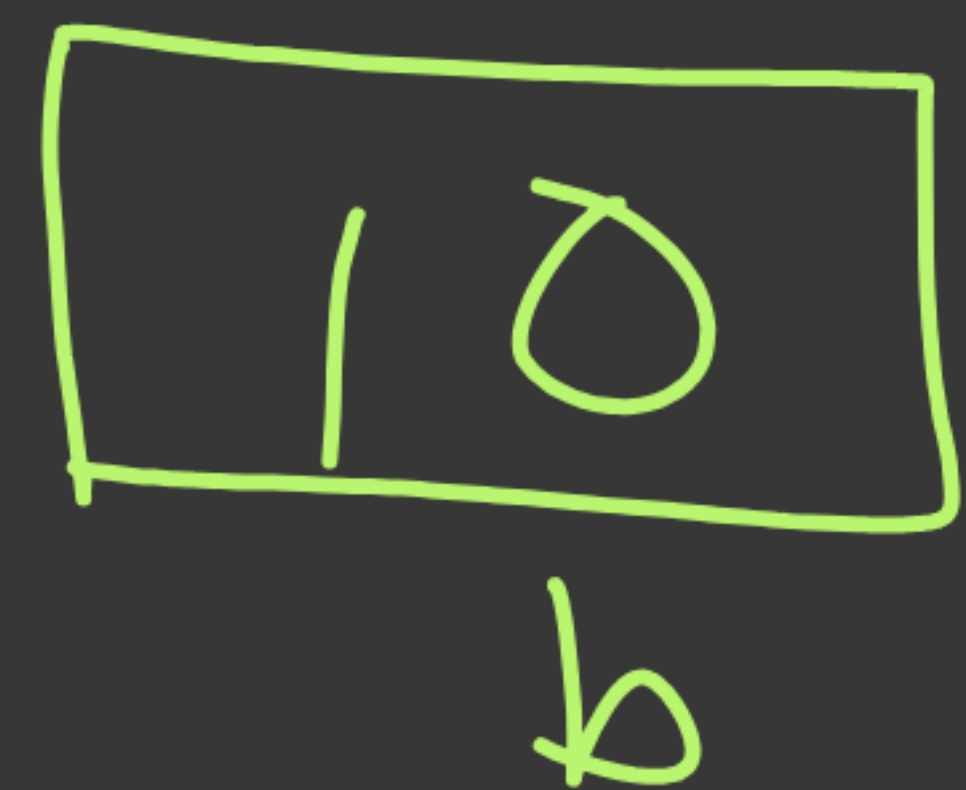
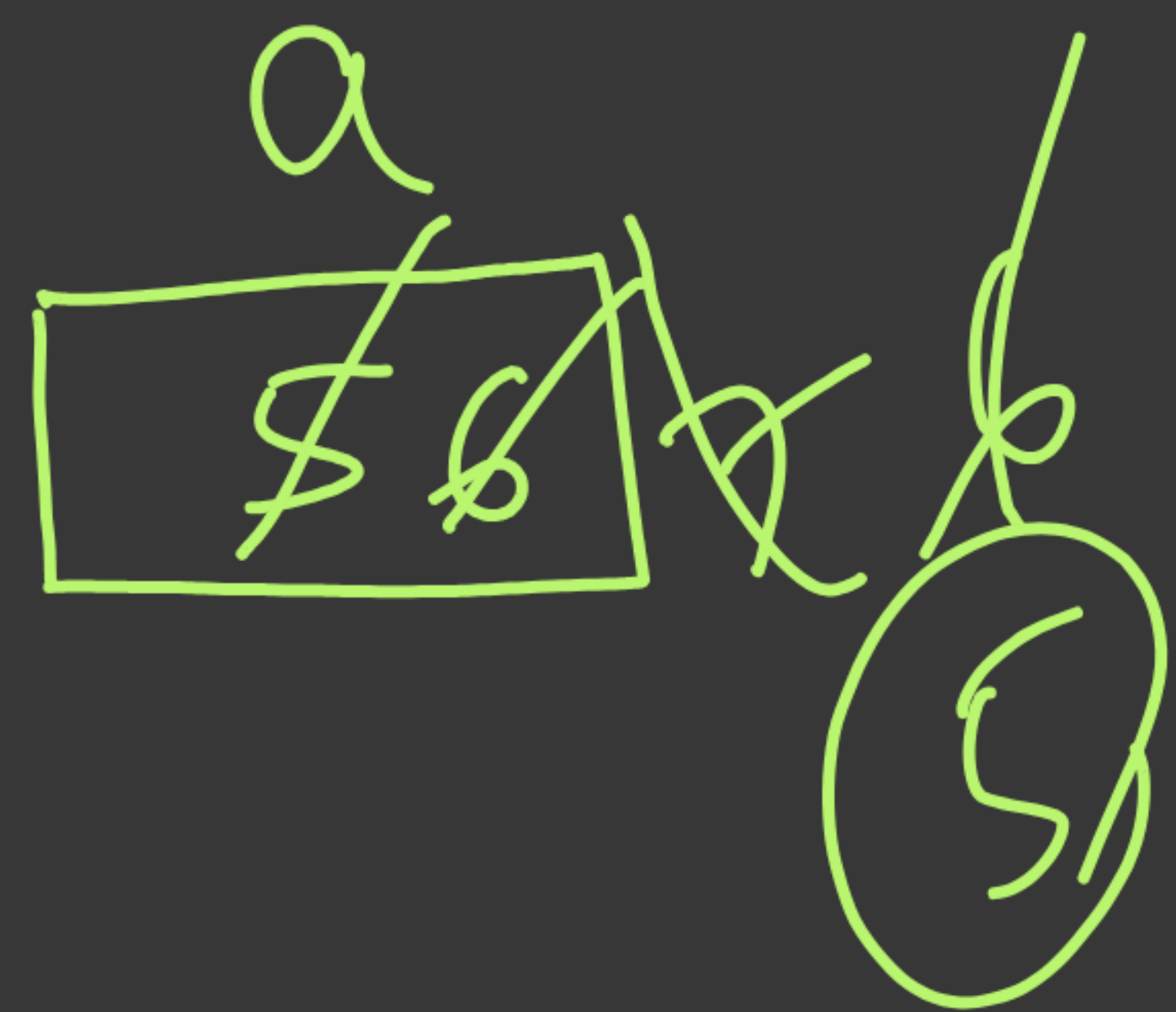
5 + 7

```

int a = 5;
int b = a++ + ++a - a;
cout << a + " " + b;

```

Annotations for the code above:
 - The initial value 5 is circled.
 - An arrow from the first 5 to the first `a++` is labeled $a = a + 1$ and $b = 5 + 1$.
 - An arrow from the second `++a` to the `a` in `++a` is labeled $a = a + 1$ and $7 = 6 + 1$.
 - The `a` in `++a` is circled with 7.
 - The `a` in `- a` is circled with 7.
 - An arrow from the circled 7 to the `a` in `- a` is labeled $a = a - 1$ and $6 = 7 - 1$.
 - The final `a` is circled with 5.
 - An arrow from the final `a` to the `a` in `a++` is labeled $a = a - 1$ and $5 = 6 - 1$.

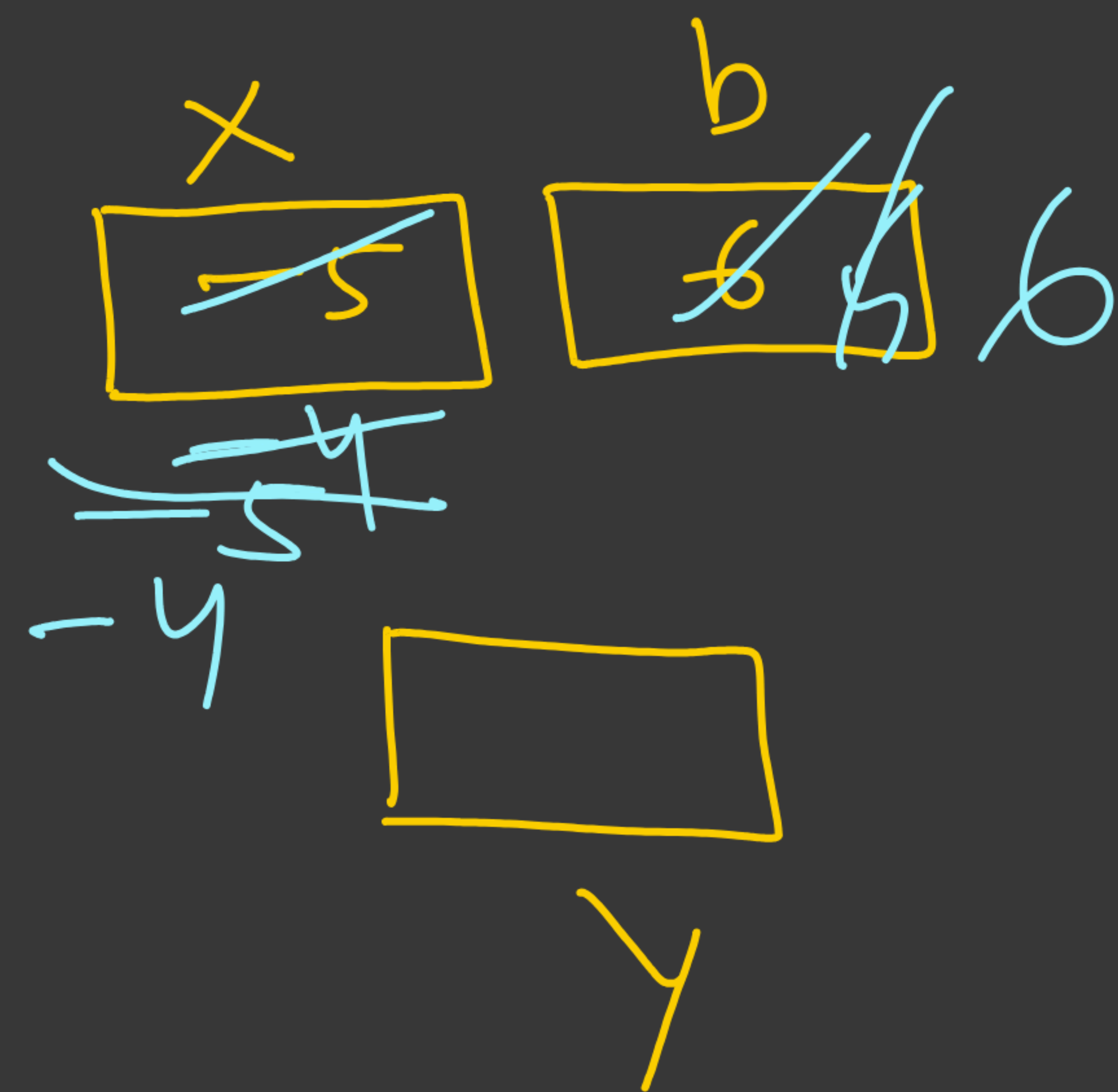


5	10
4	9
6	12
4	10
4	11

$$5 + \cancel{7} - \cancel{7} + 5 = 10$$

$\text{int } x = -5, b = 6$
 $\text{int } y = \underbrace{x++}_{-5} - \underbrace{--x}_{-5} + \underbrace{b--}_{6} - \underbrace{x++}_{-5} + \underbrace{++b}_{6};$

$\text{cout} \ll (x + " " + y + " " + b),$
 $\quad \quad \quad -4 \quad \quad \quad 17 \quad \quad \quad 6$



$$\begin{aligned}
 & -5 - (-5) + 6 - (-5) \\
 & -5 + 5 + 6 + 5 + 6
 \end{aligned}$$